

DecodeLabs Data Analytics Project 2

Exploratory Data Analysis (EDA)

Objective

The main objective of the project is to clean and transform a raw e-commerce dataset so that it becomes ready for Analysis

Specifically, the Project aims to:

- a. Understanding the structure and characteristics of the data
- b. Performing exploratory data analysis (EDA)
- c. Calculating descriptive statistics
- d. Identifying patterns and trends
- e. Detecting potential outliers
- f. Analysing product sales performance
- g. Examining relationships between numerical variables
- h. Creating visualizations to communicate findings

Import libraries

Load dataset

```
In [183... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("Cleaned_dataset.csv")
df.head()
```

Out[183...

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SA
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SA
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREI
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SA
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SA



Perform Basic Statistics on Numeric columns

In [127...

```
print("/nBasic Statistics")
print(df.describe())
```

```
/nBasic Statistics
      Quantity  UnitPrice  ItemsInCart  TotalPrice
count  1200.000000  1200.000000  1200.000000  1200.000000
mean     2.945833   356.412750     5.485000  1053.968300
std     1.407557   197.177146     2.281983   819.856558
min      1.000000    11.390000     1.000000    11.390000
25%      2.000000   186.062500     4.000000   410.520000
50%      3.000000   364.210000     5.000000   823.615000
75%      4.000000   521.570000     7.000000  1578.475000
max      5.000000   699.930000    10.000000  3456.400000
```

Print and Examine The Data columns and Data Types

In [129...

```
print("/nDataset Information:")
print(df.info())
```

```

/nDataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   OrderID                1200 non-null   object
1   Date                   1200 non-null   object
2   CustomerID             1200 non-null   object
3   Product                1200 non-null   object
4   Quantity               1200 non-null   int64
5   UnitPrice              1200 non-null   float64
6   ShippingAddress         1200 non-null   object
7   PaymentMethod           1200 non-null   object
8   OrderStatus            1200 non-null   object
9   TrackingNumber          1200 non-null   object
10  ItemsInCart             1200 non-null   int64
11  CouponCode              1200 non-null   object
12  ReferralSource          1200 non-null   object
13  TotalPrice              1200 non-null   float64
14  ShippingAddresst       1200 non-null   object
dtypes: float64(2), int64(2), object(11)
memory usage: 140.8+ KB
None

```

Count Numbers of Transactions

In [131...

```

print("Count:")
print(df["TotalPrice"].count())

```

Count:
1200

Calculate the Mean Value

In [133...

```

print("Mean:")
print(df["TotalPrice"].mean())

```

Mean:
1053.9683

Calculate the Median Value

```
In [135... print("Median:")
print(df["TotalPrice"].median())
```

```
Median:
823.615
```

Calculate the Total Sales Value

```
In [136... print("/nTotal Sales:")
print(df["TotalPrice"].sum())
```

```
/nTotal Sales:
1264761.96
```

Find The Most Sold Product

```
In [140... print("/nMost Sold Product:")
print(df["Product"].value_counts().head)
```

```
/nMost Sold Product:
<bound method NDFrame.head of Product
Printer    181
Tablet     179
Chair      178
Laptop     173
Desk       170
Monitor    163
Phone      156
Name: count, dtype: int64>
```

Find Top 5 CustomerID

```
In [143... print("/nTop 5 CustomerID:")
print(df["CustomerID"].value_counts().head())
```

```
/nTop 5 CustomerID:
CustomerID
C38840    2
C14847    2
C35852    2
C97593    2
C46651    2
Name: count, dtype: int64
```

Find Top 5 ShippingAddress

```
In [145... print("/nTop 5 ShippingAddress:")
print(df["ShippingAddress"].value_counts().head())
```

```
/nTop 5 ShippingAddress:
ShippingAddress
533 Main St      8
895 Main St      6
696 Main St      5
562 Main St      5
146 Main St      5
Name: count, dtype: int64
```

IQR-Value Base Check

```
In [147... Q1 = df["TotalPrice"].quantile(0.25)
Q3 = df["TotalPrice"].quantile(0.75)
IQR = Q3 - Q1
```

```
In [149... lower_bond = Q1 - 1.5 * IQR
upper_bond = Q3 + 1.5 * IQR
outliers = df[(df["TotalPrice"] < lower_bond) | (df["TotalPrice"] > upper_bond)]
```

8 Outliers Found

```
In [151... print("/nNumber of Outliers:")
print(len(outliers))
```

```
/nNumber of Outliers:
8
```

```
In [153... print("/nOutliers:")
print(outliers)
```

/nOutliers:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
107	ORD200107	2023-03-27	C16775	Printer	5	670.75	
326	ORD200326	2024-07-01	C65986	Laptop	5	670.48	
328	ORD200328	2023-02-28	C18404	Tablet	5	674.04	
469	ORD200469	2023-11-26	C13877	Chair	5	676.98	
632	ORD200632	2023-05-02	C67260	Laptop	5	678.16	
789	ORD200789	2023-08-17	C57276	Tablet	5	691.28	
1065	ORD201065	2023-10-30	C47778	Printer	5	666.80	
1122	ORD201122	2023-06-07	C38840	Monitor	5	678.19	

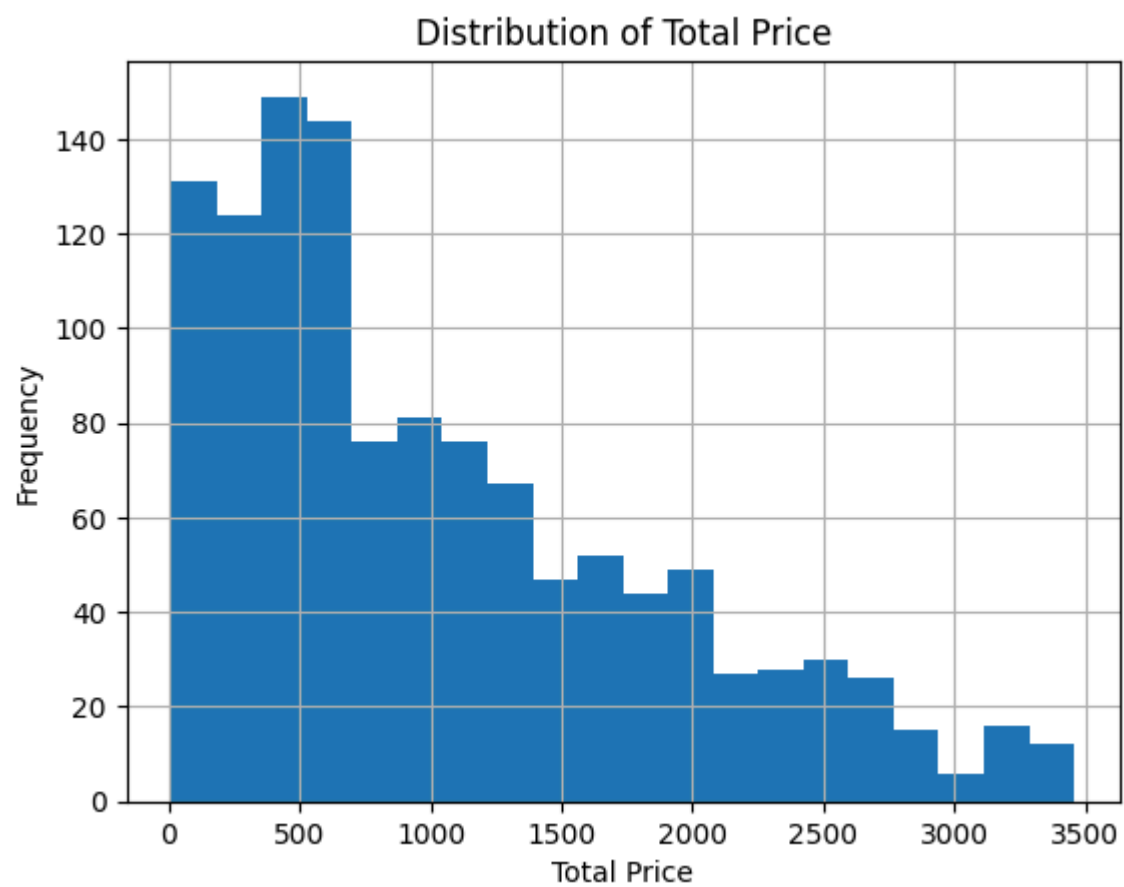
	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
107	848 Main St	Gift Card	Shipped	TRK34392124	8	
326	273 Main St	Gift Card	Returned	TRK98353867	5	
328	546 Main St	Online	Cancelled	TRK89401624	7	
469	893 Main St	Cash	Cancelled	TRK17254691	5	
632	463 Main St	Gift Card	Delivered	TRK38229104	7	
789	183 Main St	Online	Delivered	TRK75899752	10	
1065	488 Main St	Debit Card	Delivered	TRK79504329	7	
1122	766 Main St	Online	Returned	TRK32496970	8	

	CouponCode	ReferralSource	TotalPrice	ShippingAddresst
107	FREESHIP	Instagram	3353.75	848 Main St
326	SAVE10	Facebook	3352.40	273 Main St
328	SAVE10	Google	3370.20	546 Main St
469	NoCode	Facebook	3384.90	893 Main St
632	WINTER15	Facebook	3390.80	463 Main St
789	SAVE10	Email	3456.40	183 Main St
1065	SAVE10	Referral	3334.00	488 Main St
1122	NoCode	Facebook	3390.95	766 Main St

Histogram Showing The Distributions Of TotalPrice Frequency

```
In [155... plt.hist(df["TotalPrice"], bins=20)
plt.title("Distribution of Total Price")
plt.xlabel("Total Price")
plt.ylabel("Frequency")
plt.grid(True)
plt.show
```

```
Out[155... <function matplotlib.pyplot.show(close=None, block=None)>
```



Boxplot of Total Price

```
In [157... plt.figure(figsize=(8,5))
plt.boxplot(df["TotalPrice"])
plt.title("Boxplot of Total Price")
plt.ylabel("Total Price")
plt.grid(True)
plt.show
```

```
Out[157... <function matplotlib.pyplot.show(close=None, block=None)>
```



correlation Heatmap

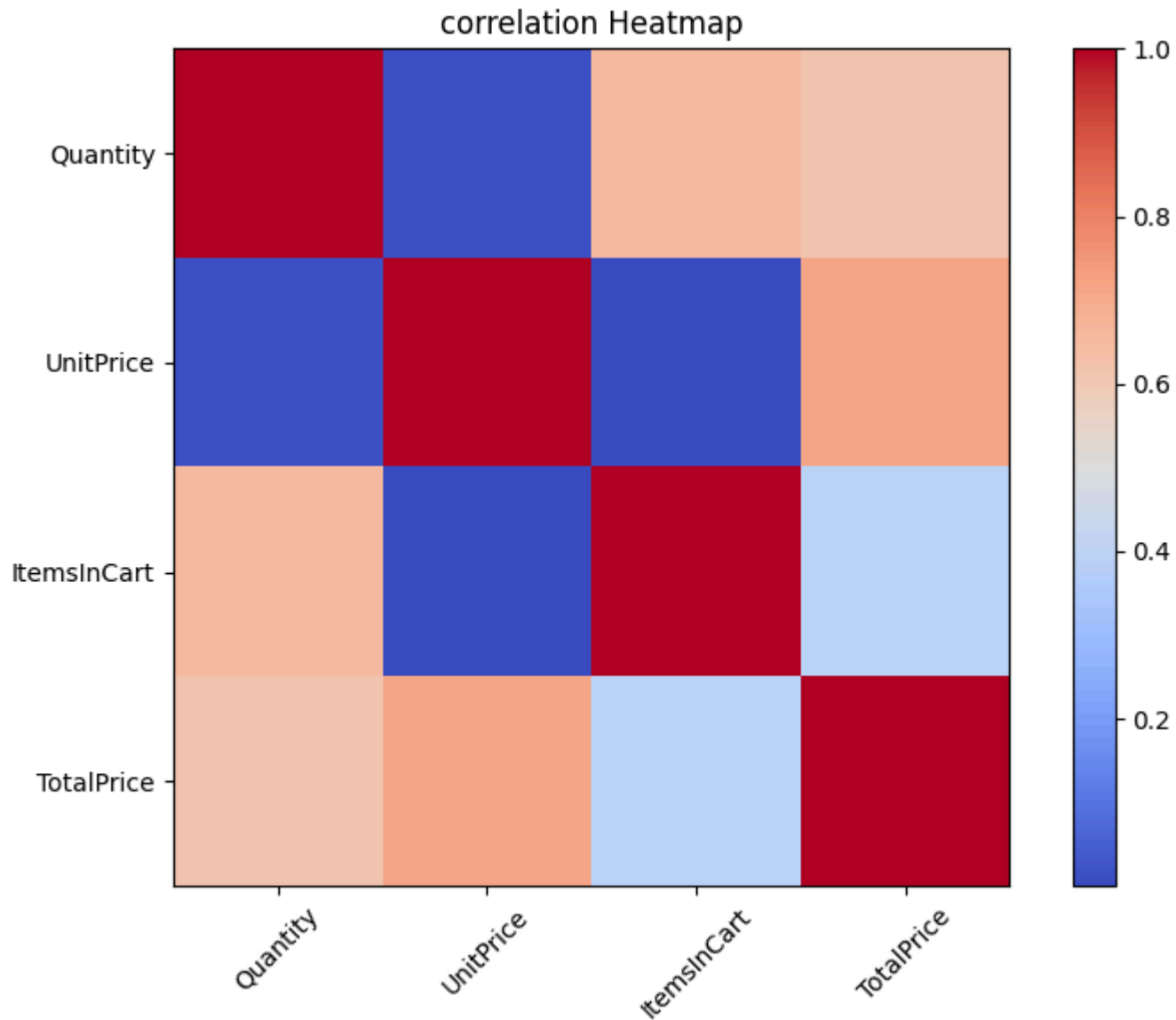
```
In [159... correlation = df.select_dtypes(include=np.number).corr()
```

```
In [161... plt.figure(figsize=(8,6))
plt.imshow(correlation, cmap="coolwarm",interpolation="nearest")
plt.colorbar()
plt.xticks(
    range(len(correlation.columns)),
    correlation.columns,
    rotation=45
)

plt.yticks(
    range(len(correlation.columns)),
    correlation.columns
)
```



```
plt.title("correlation Heatmap")
plt.tight_layout()
plt.show()
```



Top 5 Product by Sales

In [163...

```
print("/n Top 5 Product by Sales:")
sales_by_product = df.groupby("Product")['TotalPrice'].sum().reset_index()
sales_by_product
```

/n Top 5 Product by Sales:

Out[163...

	Product	TotalPrice
0	Chair	195620.11
1	Desk	167459.93
2	Laptop	192126.56
3	Monitor	175651.41
4	Phone	151722.39
5	Printer	195612.61
6	Tablet	186568.95

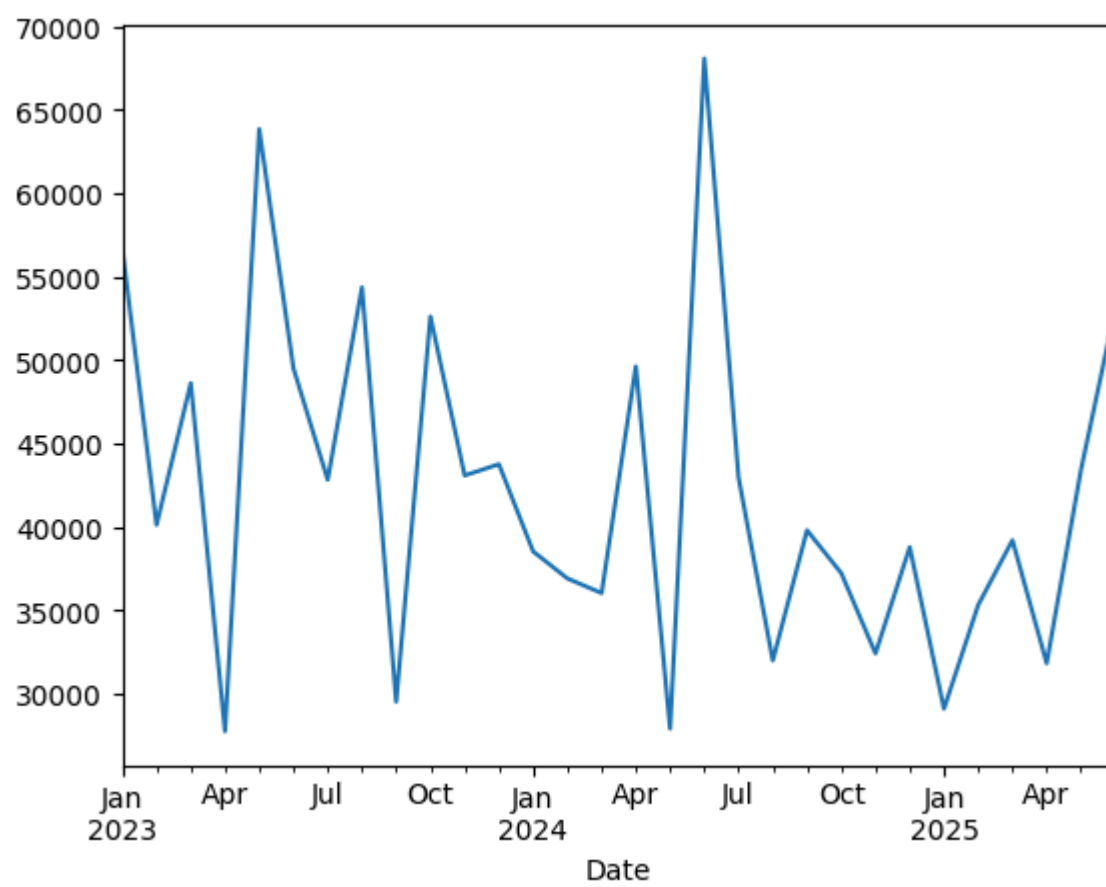
line Graph Showing The Distribution of Sales By Months And Year

In [165...

```
df['Date'] = pd.to_datetime(df['Date'])
df.groupby(df['Date'].dt.to_period('M'))['TotalPrice'].sum().plot()
```

Out[165...

<Axes: xlabel='Date'>



In [167...

```
df['Date'] = pd.to_datetime(df['Date'])
df.groupby(df['Date'].dt.to_period('M'))['TotalPrice'].sum()
```

```
Out[167... Date
2023-01      56685.75
2023-02      40117.66
2023-03      48609.37
2023-04      27751.71
2023-05      63836.84
2023-06      49500.19
2023-07      42820.66
2023-08      54352.14
2023-09      29526.67
2023-10      52607.85
2023-11      43079.67
2023-12      43754.73
2024-01      38528.08
2024-02      36909.57
2024-03      36030.90
2024-04      49613.14
2024-05      27909.11
2024-06      68068.54
2024-07      42963.98
2024-08      31991.07
2024-09      39794.98
2024-10      37226.97
2024-11      32413.76
2024-12      38785.77
2025-01      29099.40
2025-02      35317.55
2025-03      39200.66
2025-04      31821.20
2025-05      43396.64
2025-06      53047.40
Freq: M, Name: TotalPrice, dtype: float64
```

TotalPrice Group By ReferralSource

```
In [179... df.groupby('ReferralSource')['TotalPrice'].sum().reset_index()
```

Out[179...

	ReferralSource	TotalPrice
0	Email	261808.55
1	Facebook	250410.90
2	Google	250441.48
3	Instagram	275285.45
4	Referral	226815.58

OrderID Group By ReferralSource

In [175...

```
df.groupby('ReferralSource')['OrderID'].count().reset_index()
```

Out[175...

	ReferralSource	OrderID
0	Email	250
1	Facebook	228
2	Google	241
3	Instagram	259
4	Referral	222

Conclussion

This project strengthened my ability to move from raw data to meaningful insights by combining data preparation, exploratory analysis, descriptive statistics, visualization, and analytical thinking.